$$\mathcal{L} = \frac{1}{2}\, tr\left(\partial_\mu \phi\, \partial^\mu \phi\right) + g\, tr(\phi^3)$$

ϕ is an $N \times N$ matrix

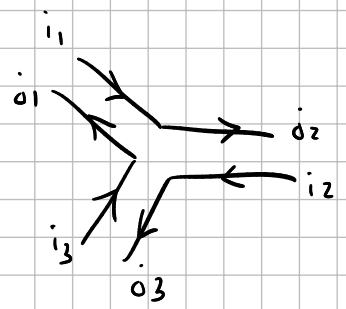
$\phi^{i}_{\ j}$ ← Colour indices

$$\mathcal{L} = \frac{1}{2}\, \underbrace{\partial_\mu \phi^{i}_{\ j}\, \partial^\mu \phi^{j}_{\ i}} + g\, \phi^{i}_{\ j}\, \phi^{j}_{\ k}\, \phi^{k}_{\ i}$$

$$\langle \phi^{i}_{\ j}(x)\, \phi^{k}_{\ l}(y) \rangle = \frac{\int D\phi\, e^{iS[\phi]}\, \phi^{i}_{\ j}(x)\, \phi^{k}_{\ l}(y)}{\int D\phi\, e^{iS[\phi]}}$$

$$\langle \phi^{i_1}_{\ j_1}(x_1)\, \phi^{i_2}_{\ j_2}(x_2) \rangle_0 \propto \delta^{i_1}_{j_2}\, \delta^{i_2}_{j_1}$$

free



$$\langle \varphi^{i_1}_{\ j_1}(p_1)\, \varphi^{i_2}_{\ j_2}(p_2)\, \varphi^{i_3}_{\ j_3}(p_3) \rangle$$

$$e^{S[\phi]} = e^{S_0[\phi]} \left(1 + \underline{g \int tr(\phi^3)} + \frac{g^2}{2}\left(\int tr(\phi^3)\right)^2 + \cdots\right)$$

$$\langle \varphi^{i_1}{}_{j_1}(p_1)\, \varphi^{i_2}{}_{j_2}(p_2)\, \varphi^{i_3}{}_{j_3}(p_3) \rangle$$

$$\parallel$$

expansion of $e^{g\int \mathrm{tr}\,\phi^3}$

$$\approx \frac{\int D\varphi\; e^{-S_0[\varphi]} \left(1 + g\int \mathrm{tr}(\phi^3)\right) \varphi^{i_1}{}_{j_1}(p_1)\, \varphi^{i_2}{}_{j_2}(p_2)\, \varphi^{i_3}{}_{j_3}(p_3)}{\int D\varphi\; e^{-S_0[\varphi]}}$$

$$\phi^{i_4}{}_{j_4}\, \phi^{i_5}{}_{j_5}\, \phi^{i_6}{}_{j_6}$$

$$\varphi^{i_4}{}_{j_4}\, \varphi^{i_5}{}_{j_5}\, \varphi^{i_6}{}_{j_6}\quad \delta^{j_6}{}_{i_4}\, \delta^{j_4}{}_{i_5}\, \delta^{j_5}{}_{i_6}$$

$$\langle \varphi^{i_1}{}_{j_1}(p_1)\, \varphi^{i_2}{}_{j_2}(p_2)\, \varphi^{i_3}{}_{j_3}(p_3) \rangle \approx$$

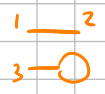
$$\langle \varphi^{i_1}{}_{j_1}(p_1)\, \varphi^{i_2}{}_{j_2}(p_2)\, \varphi^{i_3}{}_{j_3}(p_3) \int dp_4\, dp_5\, dp_6\; \mathrm{tr}\left(\phi(p_4)\,\phi(p_3)\,\phi(p_6)\right)\; \delta(p_4+p_5+p_6) \rangle_0$$

$\varphi^{i_1}{}_{j_1}(p_1)$

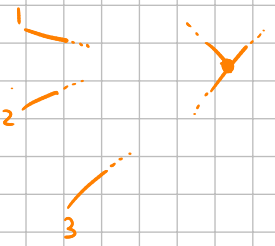
$\langle 1\,2\,3\,4\,5\,6 \rangle = \langle 1\,2\,\underbrace{3\,4}\,\underbrace{5\,6} \rangle$

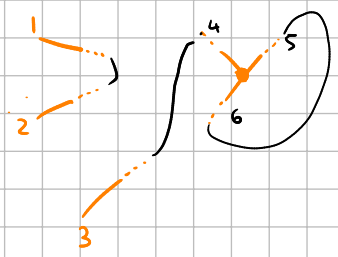
$(6-1)!!$

$$\left.\begin{array}{l} \langle 1\,2\rangle\langle 3\,4\rangle\langle 5\,6\rangle \\ \langle 1\,2\rangle\langle 3\,5\rangle\langle 4\,6\rangle \\ \langle 1\,3\rangle \quad \text{———} \quad - \end{array}\right\} \quad 15 \text{ total}$$



→ contract 4 5 6 among themselves





$\langle 1\,4\rangle\langle 2\,5\rangle\langle 3\,6\rangle \quad \delta^{j_6}{}_{i_4}\, \delta^{j_4}{}_{i_5}\, \delta^{j_5}{}_{i_6}$

$\langle 1\,4\rangle\langle 2\,6\rangle\langle 3\,5\rangle$

$\cancel{\langle 1\,5\rangle\langle 2\,4\rangle\langle 3\,6\rangle}$

$\cancel{\langle 1\,5\rangle\langle 2\,6\rangle\langle 3\,4\rangle}$

$\cancel{\langle 1\,6\rangle\langle 2\,4\rangle\langle 3\,5\rangle}$

$\cancel{\langle 1\,6\rangle\langle 2\,5\rangle\langle 3\,4\rangle}$

double counting

$$\mathrm{tr}(AB) = \sum_{i,j} A^i{}_j B^j{}_i \qquad (AB)^i{}_j = \sum_k A^i{}_k B^k{}_j$$

$$\mathrm{tr}(ABC) = \sum_{i,j,k} A^i{}_j B^j{}_k C^k{}_i$$

$A \mathbb{1} B \mathbb{1} C$

$$= \sum_{\substack{i_1 j_1 \\ i_2 j_2 \\ i_3 j_3}} A^{i_1}{}_{j_1} B^{i_2}{}_{j_2} C^{i_3}{}_{j_3} \, \delta^{j_1}_{i_2} \delta^{j_2}_{i_3} \delta^{j_3}_{i_1}$$

momentum

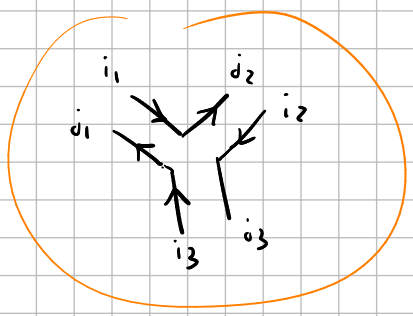
$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \int dP_4\, dP_5\, dP_6 \; \underbrace{\langle 1\,4\rangle}_{\delta^{i_1}_{j_4} \delta^{j_1}_{i_4} \delta(P_1+P_4)} \langle 2\,5\rangle \langle 3\,6\rangle \; \delta(P_4+P_5+P_6) \; \delta^{j_6}_{i_4} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6} \leftarrow \text{clear}$$

$$\sum_{i_4} \delta^{j_1}_{i_4} \delta^{j_6}_{i_4} = \delta^{j_1 j_6}$$

$$\sum_{\substack{i_4 j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_1}_{j_4} \delta^{j_1}_{i_4} \delta^{i_2}_{j_5} \delta^{j_2}_{i_5} \delta^{i_3}_{j_6} \delta^{j_3}_{i_6} \delta^{j_6}_{i_4} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6} = \delta^{i_1}_{j_2} \delta^{i_2}_{j_3} \delta^{i_3}_{j_1}$$

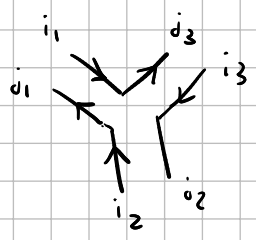
$$\parallel$$

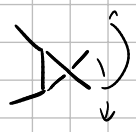
$$\sum_{\substack{j_4 \\ i_5 j_5 \\ i_6 j_6}} \delta^{i_1}_{j_4} \delta^{i_2}_{j_5} \delta^{j_2}_{i_5} \delta^{i_3}_{j_6} \delta^{j_3}_{i_6} \delta^{j_1 j_6} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6} = \sum_{\substack{j_4 \\ i_5 j_5 \\ i_6}} \delta^{i_3 j_1} \delta^{i_1}_{j_4} \delta^{i_2}_{j_5} \delta^{j_2}_{i_5} \delta^{j_3}_{i_6} \delta^{j_4}_{i_5} \delta^{j_5}_{i_6}$$



$\langle 1\,4\rangle \langle 2\,6\rangle \langle 3\,5\rangle$ non cyclic permutation of previous one swapping 2 and 3

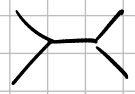
$\Rightarrow$





$$\mathcal{A}_3 = \sum_{S_n/Z_n} \text{ of amplitude with fixed ordering}$$

$n=4$ $\quad \langle \underbrace{1\ 2\ 3\ 4}_{\text{external legs}}\ \overbrace{5\ 6\ 7}^{\mathrm{tr}(\phi^3)}\ \overbrace{8\ 9\ 10}^{\mathrm{tr}(\phi^3)} \rangle$



cyclic symmetry $\quad \overrightarrow{567}\ \overleftarrow{567} \quad \overrightarrow{8910}\ \overleftarrow{8910}$

this time if we want connected, tree level things, we need to connect one of 5 6 7 with one of 8 9 10

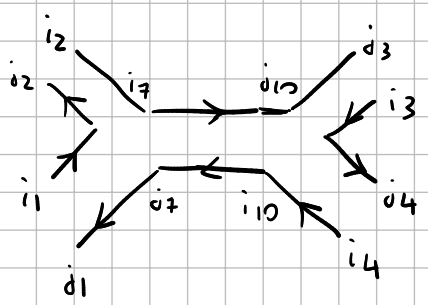
fix connection between 7 and 10

the rest needs to be contracted without loops

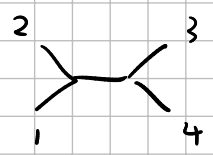
$\langle 1\ 5\rangle\langle 2\ 6\rangle\langle 3\ 8\rangle\langle 4\ 9\rangle\langle 7\ 10\rangle$ $\quad$ • permutations of 1 2 3 4

$$\langle 1\ 5\rangle\langle 2\ 6\rangle\langle 3\ 8\rangle\langle 4\ 9\rangle\langle 7\ 10\rangle\ \delta^{i_7}_{i_5}\,\delta^{i_5}_{i_6}\,\delta^{i_6}_{i_7}\ \delta^{i_{10}}_{i_8}\,\delta^{i_8}_{i_9}\,\delta^{i_9}_{i_{10}}$$

$\langle 1\ 5\rangle \to \delta^{i_1}_{i_5}\,\delta^{i_5}_{i_1}$, $\quad \langle 2\ 6\rangle \to \delta^{i_2}_{i_6}\,\delta^{i_6}_{i_2}$, $\quad \langle 4\ 9\rangle \to \delta^{i_4}_{i_9}\,\delta^{i_9}_{i_4}$, $\quad \langle 7\ 10\rangle \to \delta^{i_7}_{i_{10}}\,\delta^{i_{10}}_{i_7}$

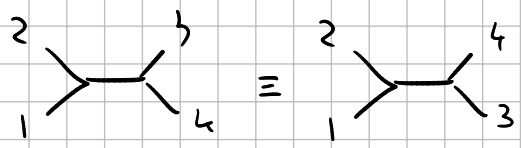


$\longleftrightarrow$



fixed colour ordering

w/o colours

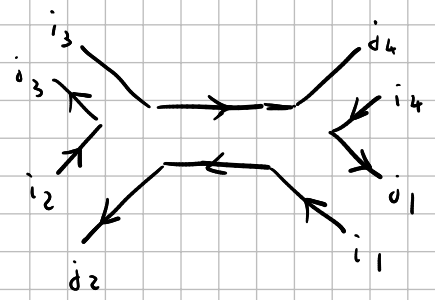


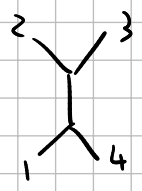
$\langle 1\,5\rangle\langle 2\,6\rangle\langle 3\,8\rangle\langle 4\,9\rangle\langle 7\,10\rangle$

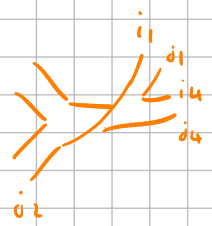
$\downarrow$ cyclic permutation

other signs $\langle 2\,5\rangle\langle 3\,6\rangle\langle 4\,8\rangle\langle 1\,9\rangle\langle 7\,10\rangle$

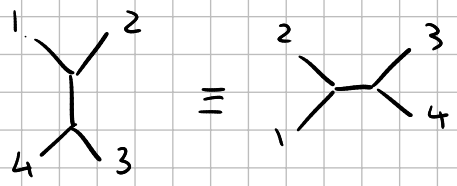
$1 \to 2$
$2 \to 3$
$3 \to 4$
$4 \to 1$







$\langle 3\,5\rangle\langle 4\,6\rangle\langle 1\,8\rangle\langle 2\,9\rangle\langle 7\,10\rangle$



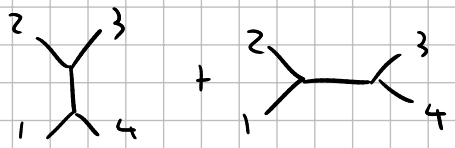
in the paper $T_1(T^{\sigma(1)}\ldots T^{\sigma(n)})$

$$\mathcal{A}_4 = \sum_{\sigma\in S_4/\mathbb{Z}_4} \overbrace{\delta^{i_1}_{o_2}\,\delta^{i_2}_{o_3}\,\delta^{i_3}_{o_4}\,\delta^{i_4}_{o_1}}\ \underbrace{A_4(\sigma(1)\ldots\sigma(4))}$$

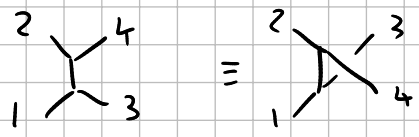
momentum amplitudes (planar)

(partial amplitude)

$A_4(1\,2\,3\,4) =$



this includes



(u-channel)

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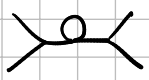
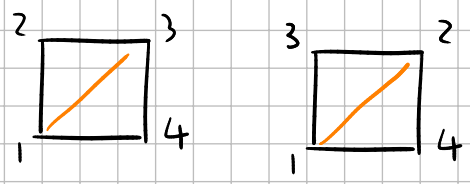
factor of N




2
3
1
4
3
2
1
4